# Convex Optimization Exercises - Sheet 5

An old landfill has contaminated the groundwater beneath it. Below the site lies a layer of impermeable clay. Long ago a river cut a valley into this clay, and the valley has since been filled with sand. Today the contaminated groundwater flows out of the site through this buried, sand-filled channel. Across the channel, the water table is level.

You have been hired to stop the contaminated water from leaving the site. There are two options and you may combine them.

**Option 1:** A wall can be build across the channel. To stop the water flow, the wall must reach from the clay at the bottom up to the water table, and it must span the full width of the saturated sand.

**Option 2:** Groundwater can be pumped out of the channel upstream of the wall, lowering the water table. The pumped water is contaminated and must be treated. You may chooses the pumping rate.

**The channel:** Without pumping, the water table sits 8 m below the surface and the surface is flat. At the wall construction site, the depth of the clay layer $d$ depends on the horizontal distance $x$ from the deepest point via the following formula, measured in meter.

$$d = 24 - x^2/2000$$

**The well:** Every cubic meter of water pumped per day lowers the water table by 4 cm. The well can be built with exactly the capacity it needs and runs continuously at that rate.

## Costs

- Wall construction cost €400 per $m^2$ of wall area.
- Pump installation costs €400 per $m^3\,d^{-1}$ of capacity.
- Water treatment costs €0.50 per $m^3$ of pumped water.
- Planning horizon: 30 years. The wall and the well are paid for at the start.

## Your task

Find the cheapest design that stops all contaminated water from leaving the site. Formulate this as an optimization problem, justify your modelling choices, show that the problem is convex, and determine the optimal pumping rate and the dimensions of the wall.